



Language: **English**

**Day 0**

---

*Friday, May 7, 2021*

**Problem 1.** For a positive integer  $n$ , consider a square cake which is divided into  $n \times n$  pieces with at most one strawberry on each piece. We say that such a cake is *delicious* if both diagonals are fully occupied, and each row and each column has an odd number of strawberries.

Find all positive integers  $n$  such that there is an  $n \times n$  delicious cake with exactly  $\left\lceil \frac{n^2}{2} \right\rceil$  strawberries on it.

**Problem 2.** Prove that, for all positive integers  $m$  and  $n$ , we have

$$\left\lfloor m\sqrt{2} \right\rfloor \cdot \left\lfloor n\sqrt{7} \right\rfloor < \left\lfloor mn\sqrt{14} \right\rfloor.$$

**Problem 3.** Let  $P$  be a point on the circumcircle of an acute-angled triangle  $ABC$ . Let  $D$ ,  $E$ , and  $F$  be the reflections of  $P$  in the perpendicular bisectors of the altitudes of triangle  $ABC$  parallel to  $BC$ ,  $CA$ , and  $AB$ , respectively. Denote by  $\omega_A$ ,  $\omega_B$ , and  $\omega_C$  the circumcircles of triangles  $ADP$ ,  $BEP$ , and  $CFP$ , respectively. Denote by  $\omega$  the circumcircle of the triangle formed by the perpendicular bisectors of segments  $AD$ ,  $BE$ , and  $CF$ .

Show that  $\omega_A$ ,  $\omega_B$ ,  $\omega_C$ , and  $\omega$  have a common point.

*Language: English*

*Time: 4 hours 30 minutes  
Each problem is worth 7 points*



Language: **English**

**Day 1**

*Friday, May 14, 2021*

**Problem 1.** Given a positive integer  $k$ , show that there exists a prime  $p$  such that one can choose distinct integers  $a_1, a_2, \dots, a_{k+3} \in \{1, 2, \dots, p-1\}$  such that  $p$  divides  $a_i a_{i+1} a_{i+2} a_{i+3} - i$  for all  $i = 1, 2, \dots, k$ .

**Problem 2.** Suppose that  $a, b, c, d$  are positive real numbers satisfying  $(a+c)(b+d) = ac + bd$ . Find the smallest possible value of

$$S = \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

**Problem 3.** Let  $p$  be an odd prime, and put  $N = \frac{1}{4}(p^3 - p) - 1$ . The numbers  $1, 2, \dots, N$  are painted arbitrarily in two colors, red and blue. For any positive integer  $n \leq N$ , denote by  $r(n)$  the fraction of integers in  $\{1, 2, \dots, n\}$  that are red.

Prove that there exists a positive integer  $a \in \{1, 2, \dots, p-1\}$  such that  $r(n) \neq a/p$  for all  $n = 1, 2, \dots, N$ .

*Language: English*

*Time: 4 hours 30 minutes  
Each problem is worth 7 points*



Language: **English**

**Day 2**

*Saturday, May 15, 2021*

**Problem 1.** Let  $ABCD$  be a convex quadrilateral with  $\angle ABC > 90^\circ$ ,  $\angle CDA > 90^\circ$ , and  $\angle DAB = \angle BCD$ . Denote by  $E$  and  $F$  the reflections of  $A$  in lines  $BC$  and  $CD$ , respectively. Suppose that segments  $AE$  and  $AF$  meet line  $BD$  at  $K$  and  $L$ , respectively. Prove that the circumcircles of triangles  $BEK$  and  $DFL$  are tangent to each other.

**Problem 2.** For any odd prime  $p$  and any integer  $n$ , let  $d_p(n) \in \{0, 1, \dots, p-1\}$  denote the remainder when  $n$  is divided by  $p$ . We say that  $(a_0, a_1, a_2, \dots)$  is a  $p$ -sequence, if  $a_0$  is a positive integer coprime to  $p$ , and  $a_{n+1} = a_n + d_p(a_n)$  for  $n \geq 0$ .

- (a) Do there exist infinitely many primes  $p$  for which there exist  $p$ -sequences  $(a_0, a_1, a_2, \dots)$  and  $(b_0, b_1, b_2, \dots)$  such that  $a_n > b_n$  for infinitely many  $n$ , and  $b_n > a_n$  for infinitely many  $n$ ?
- (b) Do there exist infinitely many primes  $p$  for which there exist  $p$ -sequences  $(a_0, a_1, a_2, \dots)$  and  $(b_0, b_1, b_2, \dots)$  such that  $a_0 < b_0$ , but  $a_n > b_n$  for all  $n \geq 1$ ?

**Problem 3.** Consider any rectangular table having finitely many rows and columns, with a real number  $a(r, c)$  in the cell in row  $r$  and column  $c$ . A pair  $(R, C)$ , where  $R$  is a set of rows and  $C$  is a set of columns, is called a *saddle pair* if the following two conditions are satisfied:

- (i) For each row  $r'$ , there is an  $r \in R$  such that  $a(r, c) \geq a(r', c)$  for all  $c \in C$ ;
- (ii) For each column  $c'$ , there is a  $c \in C$  such that  $a(r, c) \leq a(r, c')$  for all  $r \in R$ ;

A saddle pair  $(R, C)$  is called a *minimal pair* if for each saddle pair  $(R', C')$  with  $R' \subseteq R$  and  $C' \subseteq C$ , we have  $R' = R$  and  $C' = C$ .

Prove that any two minimal pairs contain the same number of rows.

Language: *English*

Time: *4 hours 30 minutes*

Each problem is worth *7 points*



Language: **English**

**Day 3**

*Tuesday, May 18, 2021*

**Problem 1.** Let  $n$  be a positive integer, and set  $N = 2^n$ . Determine the smallest real number  $a_n$  such that, for all real number  $x$ ,

$$\sqrt[N]{\frac{x^{2N} + 1}{2}} \leq a_n(x - 1)^2 + x.$$

**Problem 2.** Consider a plane containing  $n \geq 6$  pairwise disjoint disks  $D_1, D_2, \dots, D_n$  with radii  $R_1 \geq R_2 \geq \dots \geq R_n$ . For every  $i = 1, 2, \dots, n$ , a point  $P_i$  is chosen in disk  $D_i$ . Let  $O$  be an arbitrary point in the plane. Prove that

$$OP_1 + OP_2 + \dots + OP_n \geq R_6 + R_7 + \dots + R_n.$$

(A *disk* is the region bounded by a circle, including its boundary.)

**Problem 3.** Determine all functions  $f$  defined on the set of all positive integers and taking non-negative integer values, satisfying the following three conditions:

- (i)  $f(n) \neq 0$  for at least one  $n$ ;
- (ii)  $f(xy) = f(x) + f(y)$  for every positive integers  $x$  and  $y$ ;
- (iii) there are infinitely many positive integers  $n$  such that  $f(k) = f(n - k)$  for all  $k < n$ .

*Language: English*

*Time: 4 hours 30 minutes  
Each problem is worth 7 points*



Language: **English**

**Day 4**

---

Wednesday, May 19, 2021

**Problem 1.** For each prime  $p$ , there is a kingdom of  $p$ -Landia consisting of  $p$  islands numbered  $1, 2, \dots, p$ . Two distinct islands numbered  $n$  and  $m$  are connected by a bridge if and only if  $p$  divides  $(n^2 - m + 1)(m^2 - n + 1)$ . The bridges may pass over each other, but cannot cross. Prove that there are infinitely many primes  $p$  such that there are two islands in  $p$ -Landia not connected by a chain of bridges.

**Problem 2.** Let  $ABCD$  be a cyclic quadrilateral with no two sides parallel. Let  $K$ ,  $L$ ,  $M$ , and  $N$  be points lying on sides  $AB$ ,  $BC$ ,  $CD$ , and  $DA$ , respectively, such that  $KLMN$  is a rhombus with  $KL \parallel AC$  and  $LM \parallel BD$ . Let  $\omega_1$ ,  $\omega_2$ ,  $\omega_3$ , and  $\omega_4$  be the incircles of triangles  $ANK$ ,  $BKL$ ,  $CLM$ , and  $DMN$ , respectively. Prove that the two internal common tangents to  $\omega_1$  and  $\omega_3$ , and the two internal common tangents to  $\omega_2$  and  $\omega_4$  are concurrent. (An *internal common tangent* of two circles is a tangent of both circles that intersects the segment joining the centers.)

**Problem 3.** Determine all functions  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  such that

$$f^{a^2+b^2}(a+b) = af(a) + bf(b)$$

for every  $a, b \in \mathbb{Z}$ . (Here,  $f^n$  denotes the  $n^{\text{th}}$  iteration of  $f$ , i.e.  $f^0(x) = x$  and  $f^{n+1}(x) = f(f^n(x))$  for all  $n \geq 0$ .)

Language: *English*

Time: *4 hours 30 minutes*  
Each problem is worth *7 points*



Language: **English**

**Day 5**

*Friday, May 21, 2021*

**Problem 1.** In a regular 100-gon, 41 vertices are colored black and the remaining 59 vertices are colored white. Prove that there exist 24 convex quadrilaterals  $Q_1, Q_2, \dots, Q_{24}$  whose corners are vertices of the 100-gon, so that

- the quadrilaterals  $Q_1, Q_2, \dots, Q_{24}$  are pairwise disjoint, and
- every quadrilateral  $Q_i$  has three corners of one color and one corner of the other colors.

**Problem 2.** Let  $\mathcal{A}$  be the set of all  $n \in \mathbb{N}$  for which there exist  $k \in \mathbb{N}$  and  $a_0, a_1, \dots, a_{k-1} \in \{1, 2, \dots, 9\}$  such that  $a_0 \geq a_1 \geq \dots \geq a_{k-1}$  and  $n = a_0 + a_1 \cdot 10^1 + \dots + a_{k-1} \cdot 10^{k-1}$ . Let  $\mathcal{B}$  be the set of all  $m \in \mathbb{N}$  for which there exist  $\ell \in \mathbb{N}$  and  $b_0, b_1, \dots, b_{\ell-1} \in \{1, 2, \dots, 9\}$  such that  $b_0 \leq b_1 \leq \dots \leq b_{\ell-1}$  and  $m = b_0 + b_1 \cdot 10^1 + \dots + b_{\ell-1} \cdot 10^{\ell-1}$ .

- (a) Are there infinitely many  $n \in \mathcal{A}$  such that  $n^2 - 3 \in \mathcal{A}$ ?
- (b) Are there infinitely many  $m \in \mathcal{B}$  such that  $m^2 - 3 \in \mathcal{B}$ ?

**Problem 3.** A magician intends to perform the following trick. She announces a positive integer  $n$ , along with  $2n$  real numbers  $x_1 < x_2 < \dots < x_{2n}$ , to the audience. A member of the audience then secretly chooses a polynomial  $P(x)$  of degree  $n$  with real coefficients, computes the  $2n$  values  $P(x_1), P(x_2), \dots, P(x_{2n})$ , and write down these  $2n$  values on the blackboard in non-decreasing order. After that the magician announces the secret polynomial to the audience.

Can the magician find a strategy to perform such a trick?

*Language: English*

*Time: 4 hours 30 minutes  
Each problem is worth 7 points*



Language: **English**

**Day 6**

---

*Saturday, May 22, 2021*

**Problem 1.** Let  $\mathcal{A}$  denote the set of all polynomials in three variables  $x, y, z$  with integer coefficients. Let  $\mathcal{B}$  denote the subset of  $\mathcal{A}$  formed by all polynomials which can be expressed as

$$(x + y + z)P(x, y, z) + (xy + yz + zx)Q(x, y, z) + xyzR(x, y, z)$$

with  $P, Q, R \in \mathcal{A}$ . Find the smallest non-negative integer  $n$  such that  $x^i y^j z^k \in \mathcal{B}$  for all non-negative integers  $i, j, k$  satisfying  $i + j + k \geq n$ .

**Problem 2.** The Fibonacci numbers  $F_0, F_1, F_2, \dots$  are defined inductively by  $F_0 = 0$ ,  $F_1 = 1$ , and  $F_{n+1} = F_n + F_{n-1}$  for  $n \geq 1$ .

Given an integer  $n \geq 2$ , determine the smallest size of a set  $S$  of integers such that, for every  $k = 2, 3, \dots, n$ , there exist some  $x, y \in S$  such that  $x - y = F_k$ .

**Problem 3.** Let  $I$  and  $I_A$  be the incenter and the  $A$ -excenter of an acute-angled triangle  $ABC$  with  $AB < AC$ , respectively. Let the incircle meet  $BC$  at  $D$ . Line  $AD$  meets  $BI_A$  and  $CI_A$  at  $E$  and  $F$ , respectively. Prove that the circumcircles of triangles  $AID$  and  $I_AEF$  are tangent to each other.

*Language: English*

*Time: 4 hours 30 minutes  
Each problem is worth 7 points*